

Spin Hamiltonian Framework & Property Evaluation

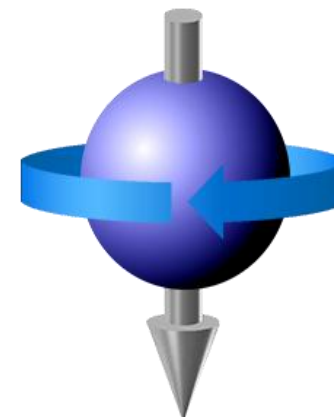
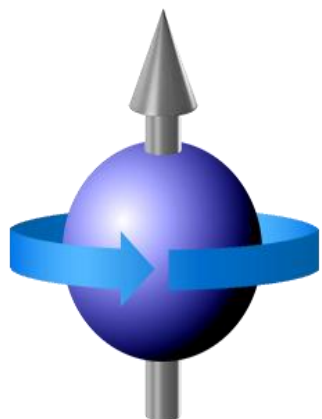
Summer Internship Presentation

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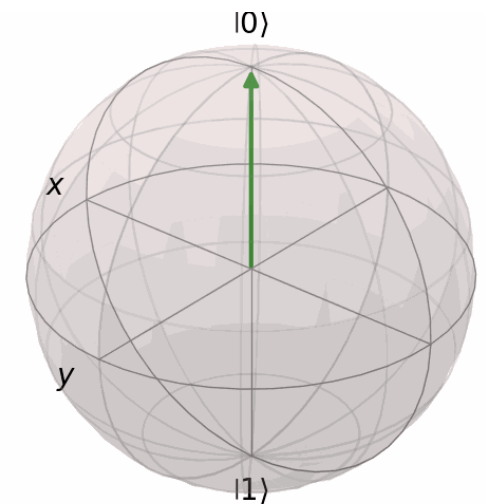
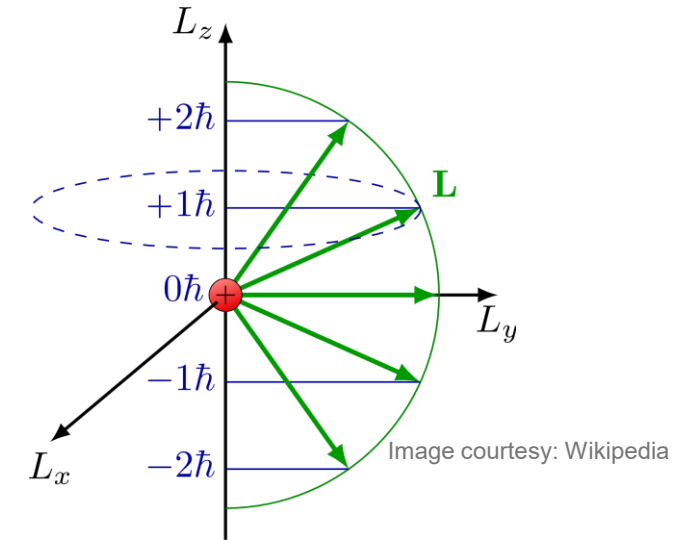
Outline

- Theory of spin
- Spin Hamiltonians
- Matrix Mechanics & Spin Algebra
- Applications of Spin Hamiltonians
- Outcome of spin properties

Introduction

Spin Angular Momentum ($\vec{S}$):

- An **intrinsic angular momentum** that particles (like electrons and protons) possess, regardless of motion
- It is *not* due to physical rotation; it is a fundamental, non-classical property, and doesn't have any classical analogue.
- Requires a unique quantum number S , (e.g, $S=1/2$ for an electron)
- For an electron, the component along z axis is given in terms of $m_s = 1/2$ spin up ($\uparrow$) and $m_s = -1/2$, spin down ($\downarrow$)



Introduction

- Spin is described by operators that satisfy the same commutation relations as a general angular momentum operator.

$$[S_x, S_y] = i\hbar S_z, \quad [S_y, S_z] = i\hbar S_x, \quad [S_z, S_x] = i\hbar S_y$$

- It should also satisfy the following eigenvalue equations in the complete set basis $\{|s, m_s\rangle\}$ such that

$$\begin{aligned} S^2 |s, m_s\rangle &= \hbar^2 s(s+1) |s, m_s\rangle \\ S_z |s, m_s\rangle &= \hbar m_s |s, m_s\rangle \end{aligned}$$

- Any general spin state can be written as a linear combination: $\psi = c_\alpha |\alpha\rangle + c_\beta |\beta\rangle$

Let $\hat{J}$ be defined as the **general angular momentum vector**.

- Each component of $\hat{J}$ can be separately diagonalized along with $\hat{J}^2$.
- $\hat{J}^2$ and any one component of $\hat{J}$ have simultaneous eigenfunctions. Since the components $\hat{J}$ do not mutually commute, we can choose only one of them to be simultaneously diagonalized with $\hat{J}^2$. By convention, choose $\hat{J}_z$.

$$\hat{J}_\pm = \hat{J}_x \pm i\hat{J}_y$$

The eigenvalue equations for $\hat{J}_+$ and $\hat{J}_-$ are given by : $J_\pm |j, m\rangle = \hbar \sqrt{j(j+1) - m(m \pm 1)} |j, m \pm 1\rangle$

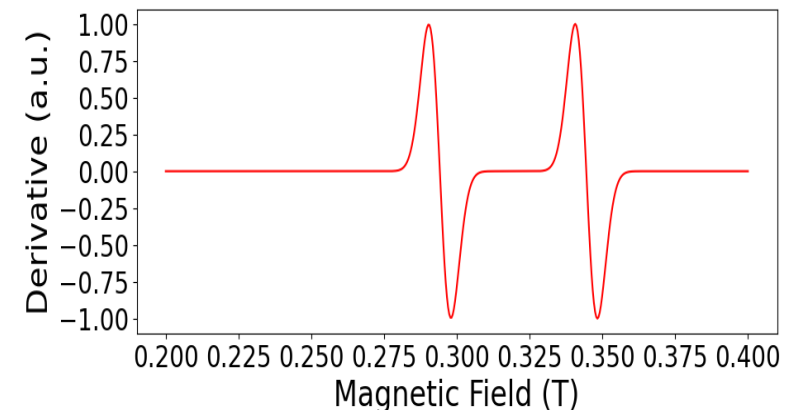
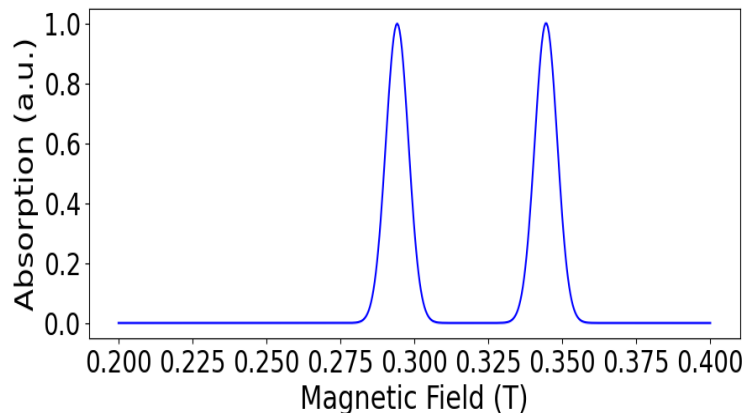
Therefore, it J^2 as well as J_z are diagonal matrices, but J_x and J_y are not.

Introduction

Where to use?

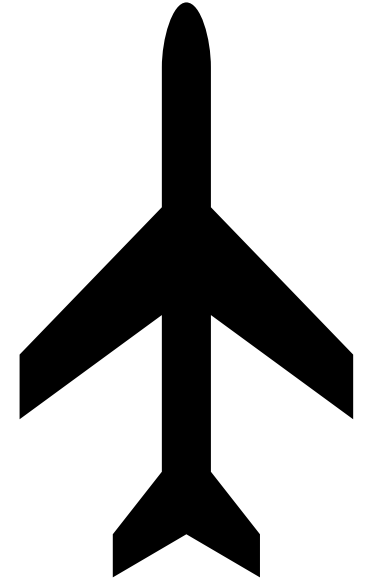
- **Very common use:** A simplified Hamiltonian taking **Zeeman** & **Hyperfine** terms is used in spectroscopy like EPR.
- It only considers the energy terms that involve the spin states of the electrons and/ or nuclei.
- It ignores kinetic energy, Coulombic interactions, etc., focusing exclusively on magnetic interactions.

$$\hat{H} = \mu_B \mathbf{B} \cdot \mathbf{g} \cdot \hat{\mathbf{S}} + \hat{\mathbf{S}} \cdot \mathbf{A} \cdot \hat{\mathbf{I}}$$



Objectives

- Code and Analyze Spin Models (in python)
- Explore Spin–Phonon Coupling
- Macroscopic Properties from Microscopic Spin Models



Results & Discussions

1. $\hat{S}_x$, $\hat{S}_y$ & $\hat{S}_z$ for any arbitrary spin (single spin centre)

```
def print_spin_operators(S):  
    hbar= 1.0  
    d = int(2 * S + 1)  
    m = np.linspace(S, -S, d)  
  
    print("m values:", m)  
    S_plus = np.zeros((d, d), dtype=complex)  
    S_z = np.zeros((d, d), dtype=complex)  
  
    for i in range(d):  
        S_z[i, i] = m[i] * hbar  
    for i in range(d - 1):  
        S_plus[i, i + 1] = hbar * np.sqrt(S * (S + 1) - m[i] * (m[i + 1]))  
  
    S_minus = S_plus.T.conj()  
  
    Sx = 0.5 * (S_plus + S_minus)  
    Sy = -0.5j * (S_plus - S_minus)  
    S2 = S * (S + 1) * np.eye(d)  
  
    print("\nSpin S = ", S)  
    print("\nSx =\n", Sx)  
    print("\nSy =\n", Sy)  
    print("\nSz =\n", S_z)  
    print("\nS^2 =\n", S2)  
  
print_spin_operators(1)
```

$$S_x = \frac{\hbar}{2} \begin{pmatrix} 0 & \sqrt{2} & 0 \\ \sqrt{2} & 0 & \sqrt{2} \\ 0 & \sqrt{2} & 0 \end{pmatrix}$$
$$S_y = \frac{\hbar}{2} \begin{pmatrix} 0 & -i\sqrt{2} & 0 \\ i\sqrt{2} & 0 & -i\sqrt{2} \\ 0 & i\sqrt{2} & 0 \end{pmatrix}$$
$$S_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$
$$S^2 = 2\hbar^2 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$



Image courtesy: Britannica

For spin $\frac{1}{2}$ systems, we introduce the so-called 'Pauli matrices'

Results & Discussions

2. Total Spin of Two Particles

$$\begin{aligned}
 (\mathbf{S}_1 + \mathbf{S}_2)^2 &= S_1^2 + S_2^2 + 2\mathbf{S}_1 \cdot \mathbf{S}_2 \\
 &= S_1^2 + S_2^2 + 2(S_{1x}S_{2x} + S_{1y}S_{2y} + S_{1z}S_{2z}) \\
 &= S_1^2 + S_2^2 + 2S_{1z}S_{2z} + S_{1+}S_{2-} + S_{1-}S_{2+}
 \end{aligned}$$

When we have **two spin 1/2 particles**,

$|\psi\rangle = |\text{total spin state}\rangle$

$$S_{\text{tot}} = |S_1 + S_2| \dots |S_1 - S_2| = 1, 0$$

Each have 3 components ($m_s = +1, 0, -1$)

S_{tot}^2	$ \uparrow\uparrow\rangle$	$ \uparrow\downarrow\rangle$	$ \downarrow\uparrow\rangle$	$ \downarrow\downarrow\rangle$
$\langle\uparrow\uparrow $	$2\hbar^2$	0	0	0
$\langle\uparrow\downarrow $	0	$\hbar^2$	$\hbar^2$	0
$\langle\downarrow\uparrow $	0	$\hbar^2$	$\hbar^2$	0
$\langle\downarrow\downarrow $	0	0	0	$2\hbar^2$

$$S_{\text{tot}}^2 = \begin{pmatrix} 2\hbar^2 & 0 & 0 & 0 \\ 0 & \hbar^2 & \hbar^2 & 0 \\ 0 & \hbar^2 & \hbar^2 & 0 \\ 0 & 0 & 0 & 2\hbar^2 \end{pmatrix}$$

$$S_+ |\uparrow\rangle = 0,$$

$$S_- |\downarrow\rangle = 0,$$

$$S_+ |\downarrow\rangle = \hbar |\uparrow\rangle,$$

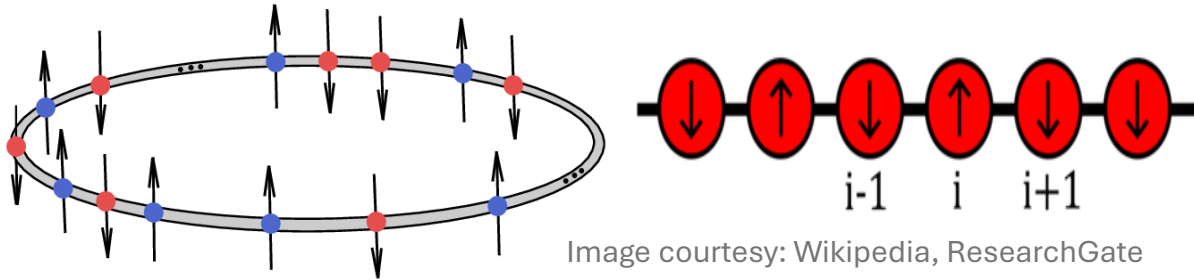
$$S_- |\uparrow\rangle = \hbar |\downarrow\rangle.$$

$$\text{Diagonalising: } S_{\text{tot}}^2 \Rightarrow \begin{cases} \text{Eigenvalues: } 0, 2\hbar^2, 2\hbar^2, 2\hbar^2 \\ \text{Singlet state: } \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle) \quad (S=0) \\ \text{Triplet states: } \begin{cases} |\uparrow\uparrow\rangle \\ \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle) \\ |\downarrow\downarrow\rangle \end{cases} \quad (S=1) \end{cases}$$

Results & Discussions

3. The Heisenberg Hamiltonian Model

$$H = J \mathbf{S}_1 \cdot \mathbf{S}_2 = J (S_{1x}S_{2x} + S_{1y}S_{2y} + S_{1z}S_{2z})$$



```
S1x = np.kron(Sx, I)
S1y = np.kron(Sy, I)
S1z = np.kron(Sz, I)

S2x = np.kron(I, Sx)
S2y = np.kron(I, Sy)
S2z = np.kron(I, Sz)

basis = ["|↑↑", "|↑↓", "|↓↑", "|↓↓"]
vector_label = []
styles = ['-', '--', ':', '-.-']
```

```
def heisenberg_hamiltonian(J):
    return J * (S1x @ S2x + S1y @ S2y + S1z @ S2z)

print("\n==== HEISENBERG MODEL =====")
J_vals = np.linspace(-1, 1, 5)
energy_levels = []

for j in J_vals:

    H = heisenberg_hamiltonian(j)
    eigvals, eigvecs = np.linalg.eigh(H)
    energy_levels.append(eigvals)

    if j in (J_vals[0], J_vals[-1], 0):
        print(f"\nJ = {j:.2f}")

    for i in range(4):
        vec = eigvecs[:, i]

        if j in (J_vals[0], J_vals[-1], 0):
            print(f"Energy Level {i+1}: {eigvals[i]:.3f} ;", end="")
            print(" Eigenvector : ", end="")
            l=""
            for (amp, state) in zip(vec, basis):
                if abs(amp) > 0.0001:
                    l= l+ f" {np.real(amp):+0.3f} {state}"
            print(l)
```

Results & Discussions

===== HEISENBERG MODEL =====

$J = -1.00$

Energy Level 1: -0.250 ; Eigenvector : +1.000 $|\uparrow\uparrow\rangle$
 Energy Level 2: -0.250 ; Eigenvector : -0.707 $|\uparrow\downarrow\rangle$ -0.707 $|\downarrow\uparrow\rangle$
 Energy Level 3: -0.250 ; Eigenvector : +1.000 $|\downarrow\downarrow\rangle$
 Energy Level 4: 0.750 ; Eigenvector : -0.707 $|\uparrow\downarrow\rangle$ +0.707 $|\downarrow\uparrow\rangle$

$J = -ve$: The triplet states get down (more stabilized), whereas singlet state go up.
FERROMAGNETIC COUPLING

$J = 0.00$

Energy Level 1: 0.000 ; Eigenvector : +1.000 $|\uparrow\uparrow\rangle$
 Energy Level 2: -0.000 ; Eigenvector : +1.000 $|\uparrow\downarrow\rangle$
 Energy Level 3: -0.000 ; Eigenvector : +1.000 $|\downarrow\downarrow\rangle$
 Energy Level 4: 0.000 ; Eigenvector : +1.000 $|\downarrow\uparrow\rangle$

$B=0$: All states are degenerate.

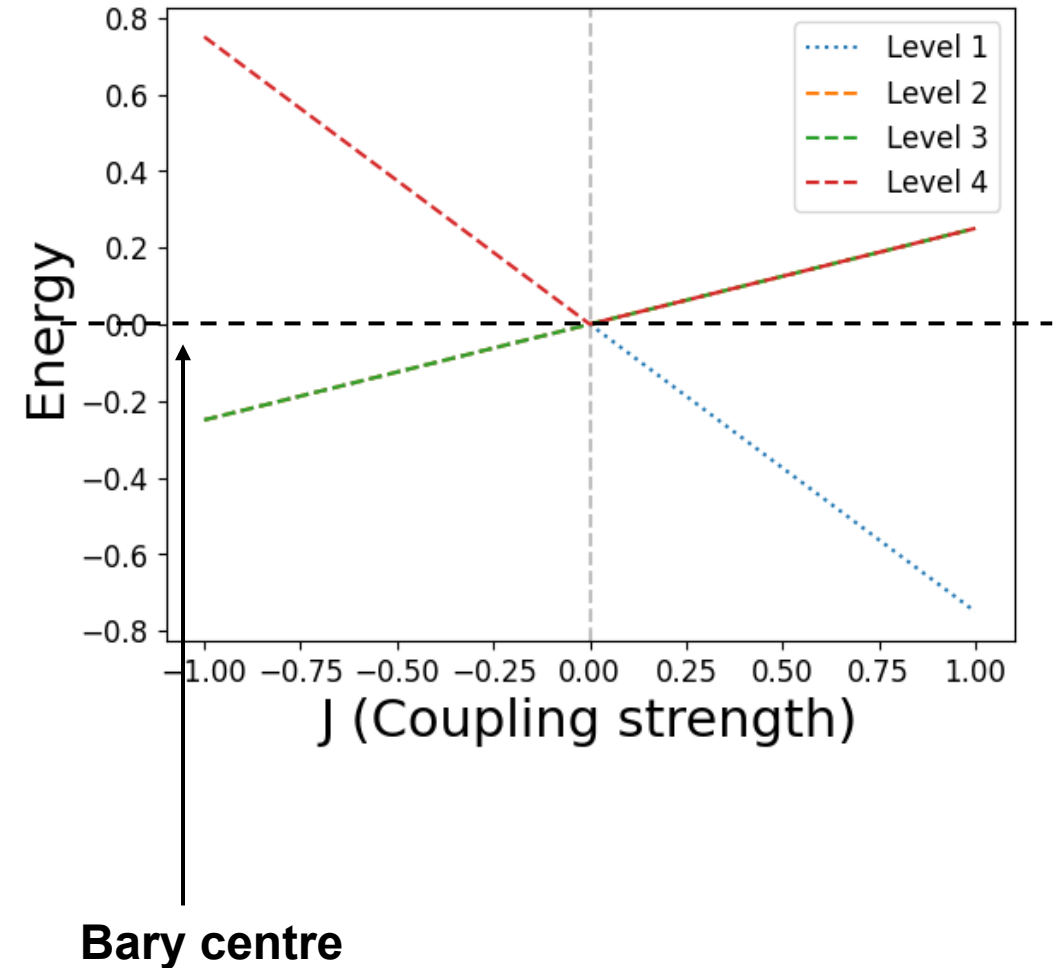
$J = 1.00$

Energy Level 1: -0.750 ; Eigenvector : +0.707 $|\uparrow\downarrow\rangle$ -0.707 $|\downarrow\uparrow\rangle$
 Energy Level 2: 0.250 ; Eigenvector : +1.000 $|\uparrow\uparrow\rangle$
 Energy Level 3: 0.250 ; Eigenvector : +0.707 $|\uparrow\downarrow\rangle$ +0.707 $|\downarrow\uparrow\rangle$
 Energy Level 4: 0.250 ; Eigenvector : +1.000 $|\downarrow\downarrow\rangle$

ANTIFERROMAGNETIC COUPLING

Eigenvalues summary :

```
[[-0.25 -0.25 -0.25  0.75 ]
 [-0.125 -0.125 -0.125  0.375]
 [ 0.    -0.    -0.    0.   ]
 [-0.375  0.125  0.125  0.125]
 [-0.75  0.25  0.25  0.25 ]]
```



Results & Discussions

4. Coupled Spin-Phonon System

$$\hat{H} = \frac{1}{2}\hbar\omega_0\sigma_z + \hbar\omega\hat{a}^\dagger\hat{a} + g\sigma_z(\hat{a} + \hat{a}^\dagger)$$

Spin-Phonon Hamiltonian (N=2 phonons, spin-1/2):

Components of the total Hamiltonian:

$$\hat{H}_{\text{spin}} = \begin{pmatrix} \frac{\hbar\omega_0}{2} & 0 & 0 & 0 \\ 0 & \frac{\hbar\omega_0}{2} & 0 & 0 \\ 0 & 0 & -\frac{\hbar\omega_0}{2} & 0 \\ 0 & 0 & 0 & -\frac{\hbar\omega_0}{2} \end{pmatrix}$$

$$\hat{H}_{\text{phonon}} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & \hbar\omega & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \hbar\omega \end{pmatrix}$$

$$\hat{H}_{\text{int}} = \begin{pmatrix} 0 & g & 0 & 0 \\ g & 0 & 0 & 0 \\ 0 & 0 & 0 & -g \\ 0 & 0 & -g & 0 \end{pmatrix}$$

Total Hamiltonian:

$$\hat{H} = \begin{pmatrix} \frac{\hbar\omega_0}{2} & g & 0 & 0 \\ g & \frac{\hbar\omega_0}{2} + \hbar\omega & 0 & 0 \\ 0 & 0 & -\frac{\hbar\omega_0}{2} & -g \\ 0 & 0 & -g & -\frac{\hbar\omega_0}{2} + \hbar\omega \end{pmatrix}$$

```
sigz = np.array([[1, 0], [0, -1]], dtype=complex)
I_spin = np.eye(2, dtype=complex)

hbar = 1.0
omega = 1.0
omega0 = 1.0
g = 1.0
N = 2

def phonon():
    a = np.zeros((N, N), dtype=complex)
    for n in range(1, N):
        a[n-1, n] = np.sqrt(n)
    return a

a = phonon()
adag = a.T.conj()
I_phonon = np.eye(N, dtype=complex)

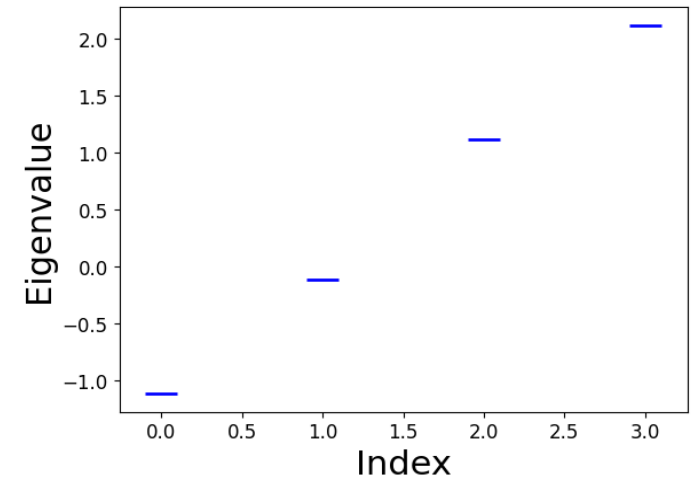
H_spin = 0.5 * hbar * omega0 * np.kron(sigz, I_phonon)
H_phonon = hbar * omega * np.kron(I_spin, adag @ a)
H_int = g * np.kron(sigz, (adag + a))

H = H_spin + H_phonon + H_int
print(np.real(H))

eigvals, eigvecs = np.linalg.eigh(H)
```

hbar = 1.0
omega = 1.0
omega0 = 1.0
g = 1.0
N = 2

$\begin{bmatrix} 0.5 & 1. & 0. & 0. \\ 1. & 1.5 & 0. & 0. \\ 0. & 0. & -0.5 & -1. \\ 0. & 0. & -1. & 0.5 \end{bmatrix}$



Results & Discussions

5. Macroscopic property from spin model: Average Magnetization

The Hamiltonian for two spin- $\frac{1}{2}$ particles with Heisenberg interaction in an external magnetic field is given by:

$$H = J \vec{S}_1 \cdot \vec{S}_2 + \mu_B B (S_{1z} + S_{2z})$$

The eigenstates are classified as: Triplet states: $S_{\text{tot}} = 1$, $E_T = \frac{J}{4} + \mu_B B M_S$, $M_S = +1, 0, -1$

$$\text{Singlet state: } S_{\text{tot}} = 0, \quad E_S = -\frac{3J}{4}$$

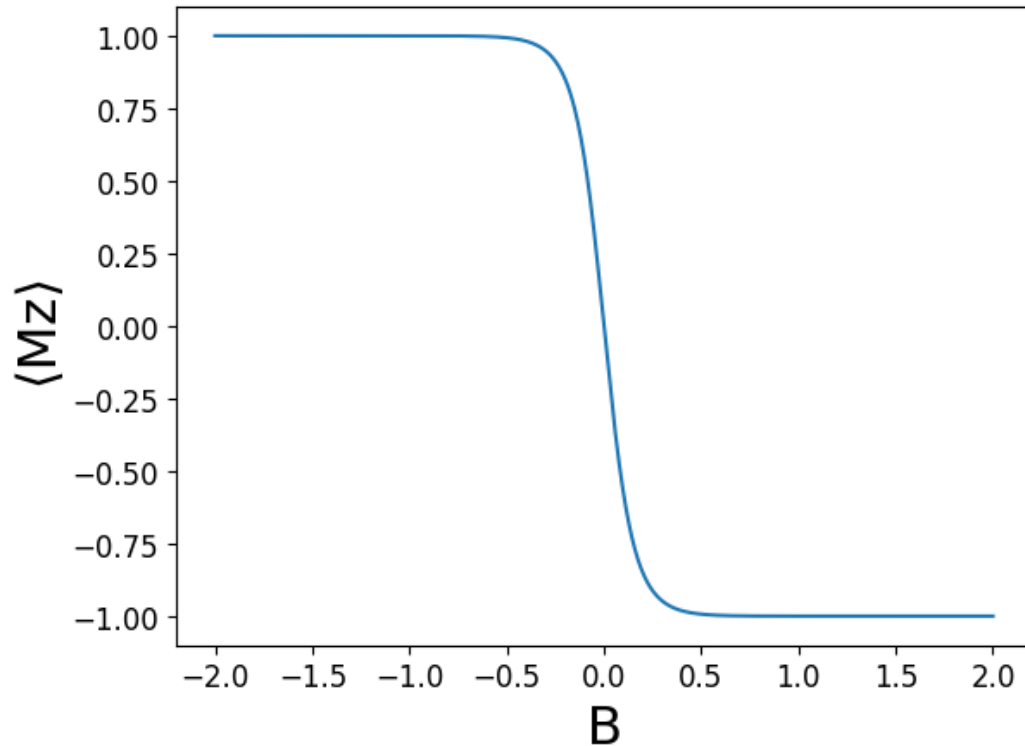
At low temperature or under a strong magnetic field, spins tend to align with the field, producing maximum magnetization. For antiferromagnetic coupling, the interaction favours opposite spin alignment, leading to suppression in net magnetization.

```
def magnetization(eigenvalues, eigenvectors):  
    Mz_op = S1z + S2z  
    w = np.exp(-eigenvalues / (kB * T)) # Array of pf components  
    Z = np.sum(w) # Partition function calculation  
    mz_values = (eigenvectors.T @ Mz_op @ eigenvectors)  
    return np.sum(mz_values * w) / Z
```

```
Mz_val = []  
for B in B_val:  
    H = J * H0 + mu * B * (S1z + S2z)  
    eigenvalues, eigenvectors = np.linalg.eigh(H) # Eigenvalues  
    Mz_val.append(magnetization(eigenvalues, eigenvectors))  
  
plt.plot(B_val, Mz_val)  
plt.xlabel("B")  
plt.ylabel("<Mz>")  
plt.title("<Mz> vs B")  
plt.grid()  
plt.show()
```

Results & Discussions

$$H = J \vec{S}_1 \cdot \vec{S}_2 + \mu_B B (S_{1z} + S_{2z})$$

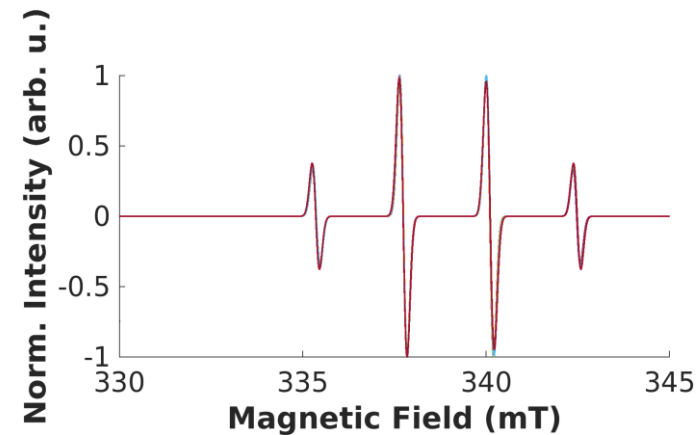
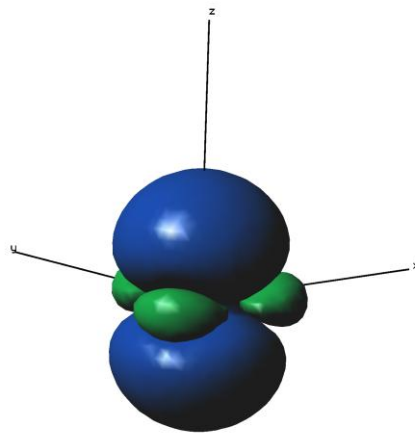


$J = -1$, $T = 0.1$ (units of thermal energy)

B	$\langle M_z \rangle$	Comments
Very large -ve value	1	Spins are aligned along the field.
Near 0	Changes rapidly	Transition region where spin alignment flips, known as the <u>spin reversal region</u> .
Very large +ve value	-1	Spins are aligned opposite to the initial direction.

Conclusions & Future Scope

- Spin Hamiltonians provide a compact framework to **model spin interactions** and **energy levels**
 - Thermal averaging links theory to experimental observables (magnetization, EPR spectra)
 - It serves as an essential tool in the field of molecular magnetism
- Currently working on PTEMPO radical-- molecular qubits; working on **decoherence** time of **molecular qubits**



Simulated in ORCA

Acknowledgements

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- Thankful to my family & friends, and to all our lab members for their support!



Courtesy: Tenor

Question
Time
Now!



Image courtesy: Moromi Nath

Thankful to all the panel members for your time and patience!